Flow Measurement

Flow

- Flow measurement important in industrial applications
- Cost of liquids and gases based on flow through pipeline
- Flow rate
  - Volume of fluid passing a point in a given time
  - Measured in litre per minute
- Total
  - Volume of liquid flowing over a period of time
- Viscosity
  - Property of liquid that measures its resistance to motion/flow



www.rshydro.co.uk/ultrasonic-flowmeter.shtml

Flow Measurement

- Up until now, we have mainly focused on measurement techniques and measurands which are directly accessible as physical variables

- Many different techniques are used to measure flow depending on the application



www.rshydro.co.uk/ultrasonic-flowmeter.shtml

Introduction to Flow Measurement

- It is easy to visualise flow is in terms of volume rates i.e. ℓ/s or m^3/s
- Most flow measurement is used in chemical processes where mass is the variable which needs to be measured
- Mass and volume are related by density:

$$\rho = \frac{m}{V}$$

- Density fluctuates with temperature and pressure changes and therefore is not a reliable indicator of mass flow if high accuracies are required



www.katronic.co.uk/prod_02.html

Introduction to Flow Measurement

- Volumetric flow is related to actual fluid movement in terms of the pipe-cross section:

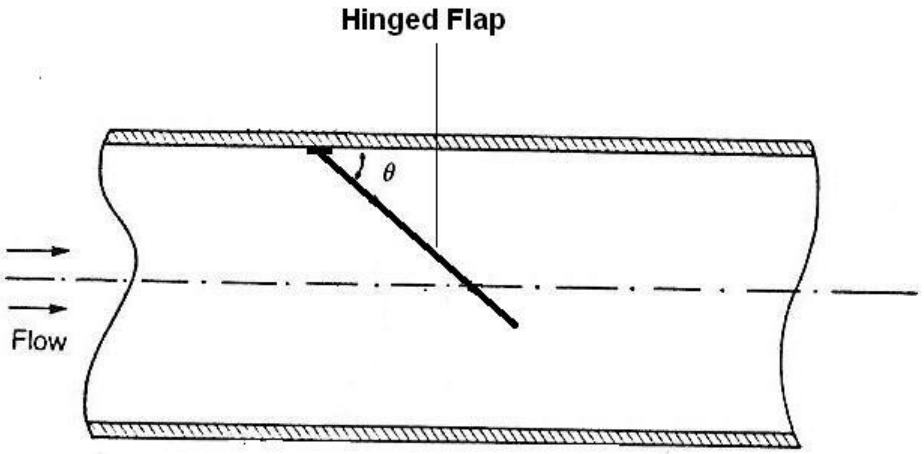
$$Q = Av$$

- Where Q is the flow rate, A is the cross sectional area and v is the flow velocity

- How do we measure flow in pipes? There are several methods, all of which have good and bad points

Flow Measurement Methods: Flow Gate

- Consists of a hinged flap which is set up to impede the flow
- The stronger the flow the more the gate will be pushed up
- The angle of the flow gate can then be measured and translated into an electrical signal



Flow Measurement Methods:
Orifice Plate

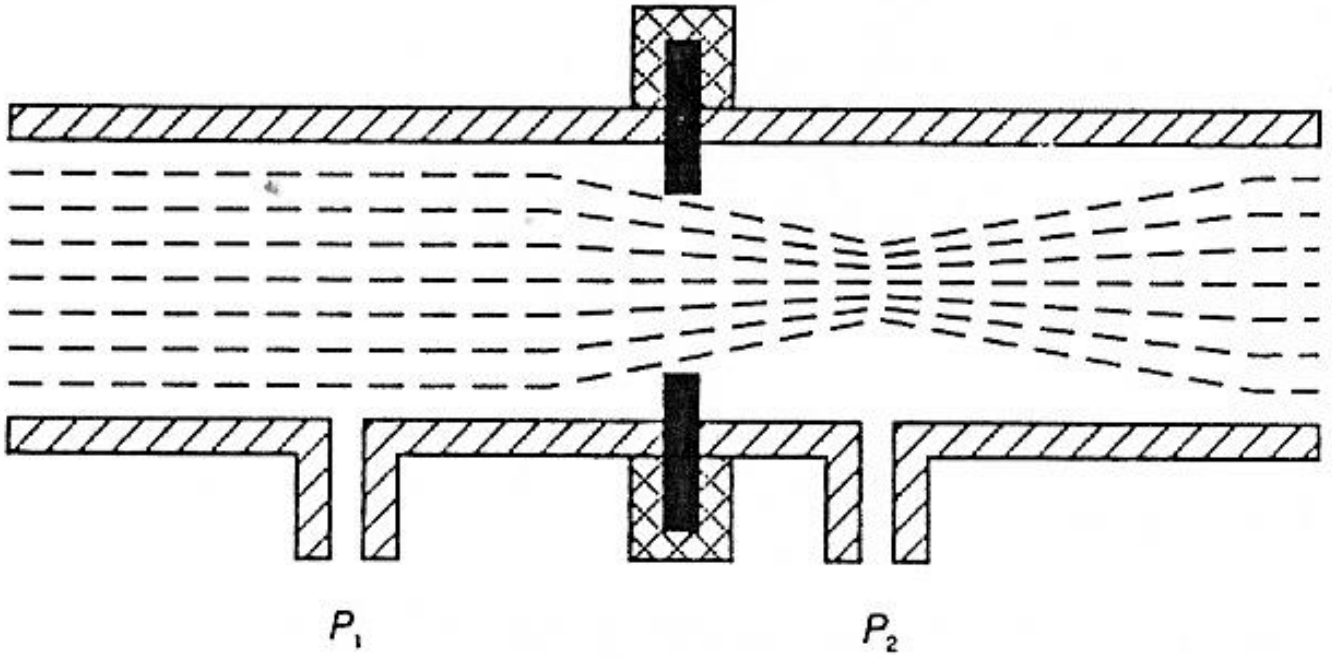
- An orifice plate is a constriction in the pipe
- The constriction causes a pressure drop in the pipe from upstream to downstream sides
- The pressure drop is related by the following expression:

$$Q = AK\sqrt{2g\Delta P}$$

- Where *g* is the gravitational constant, *ΔP* is the pressure drop, and *K* is the discharge coefficient
- *K* depends on the construction of the plate, i.e. whether it is a smooth venturi-type construction or simply a blockage plate with a hole in it.
- It also depends on the density and viscosity of the flow liquid. *K* values can be found in engineering handbooks

Flow Measurement Methods:

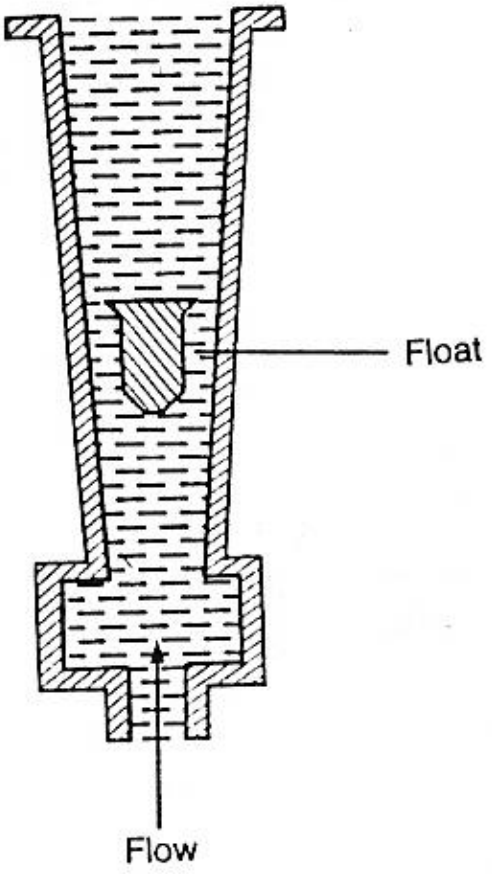
Orifice Plate



Flow Measurement Methods:

Variable Area Flow Meter

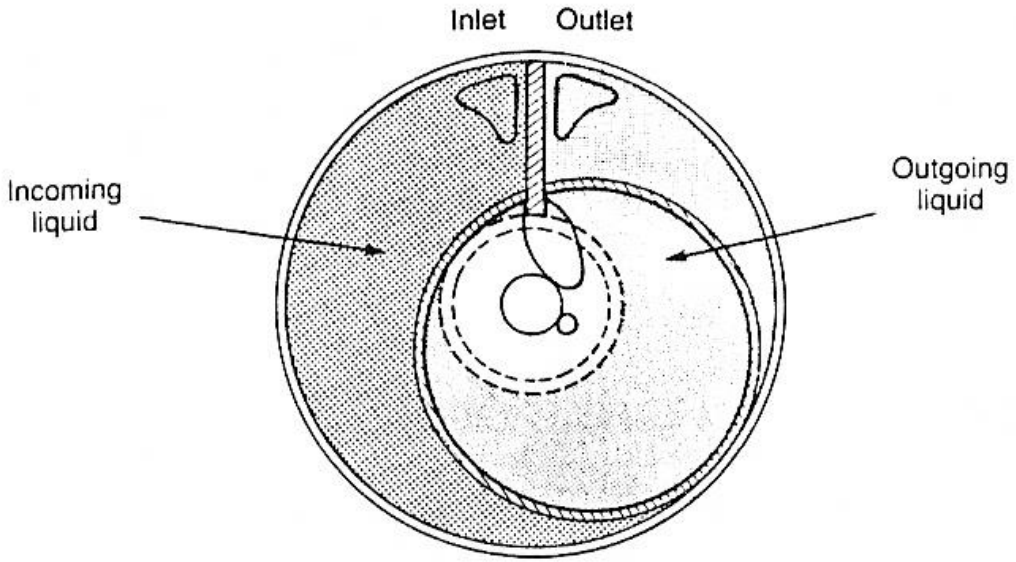
- A variable area flow meter is similar in principle to a flow gate
- Position of the float in the meter gives an indication of the flow rate
- This type of meter is always orientated vertically
- The float is pulled down by gravity while the flow pushes it up
- It comes to rest when the differential pressures on the top and bottom of the float balance out the weight of the float



Flow Measurement Methods:
Positive Displacement Flow meters

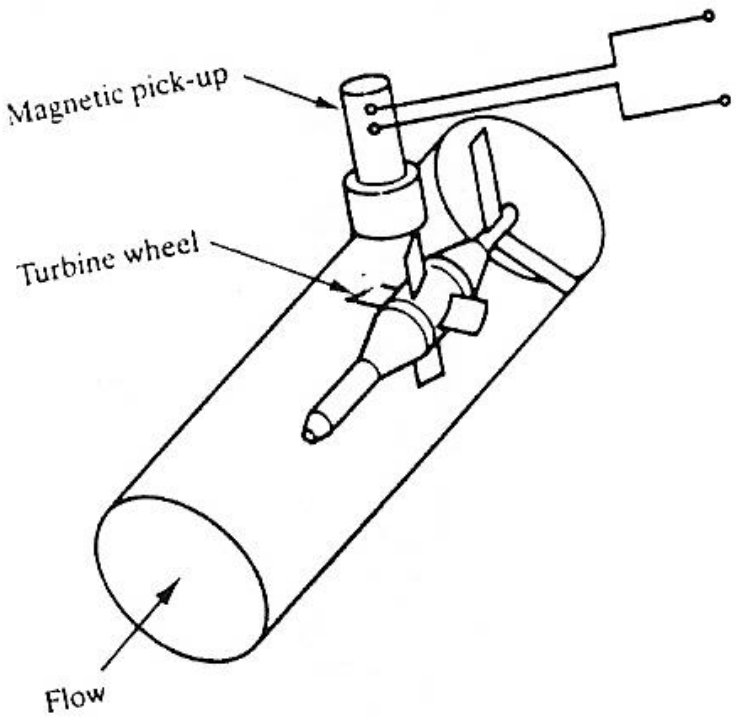
- It is a mechanical device that parcels liquid up into discrete volumes and measures the rate at which the parcels need to be processed so as to maintain the flow

- Generally the most accurate flow meters



Flow Measurement Methods:
Turbine Flow meters

- Consist of a small vaned wheel (like a propeller) in the flow path
- The propeller spin according to how fast the fluid flows past it
- The rate of rotation will give us an indication of the flow rate

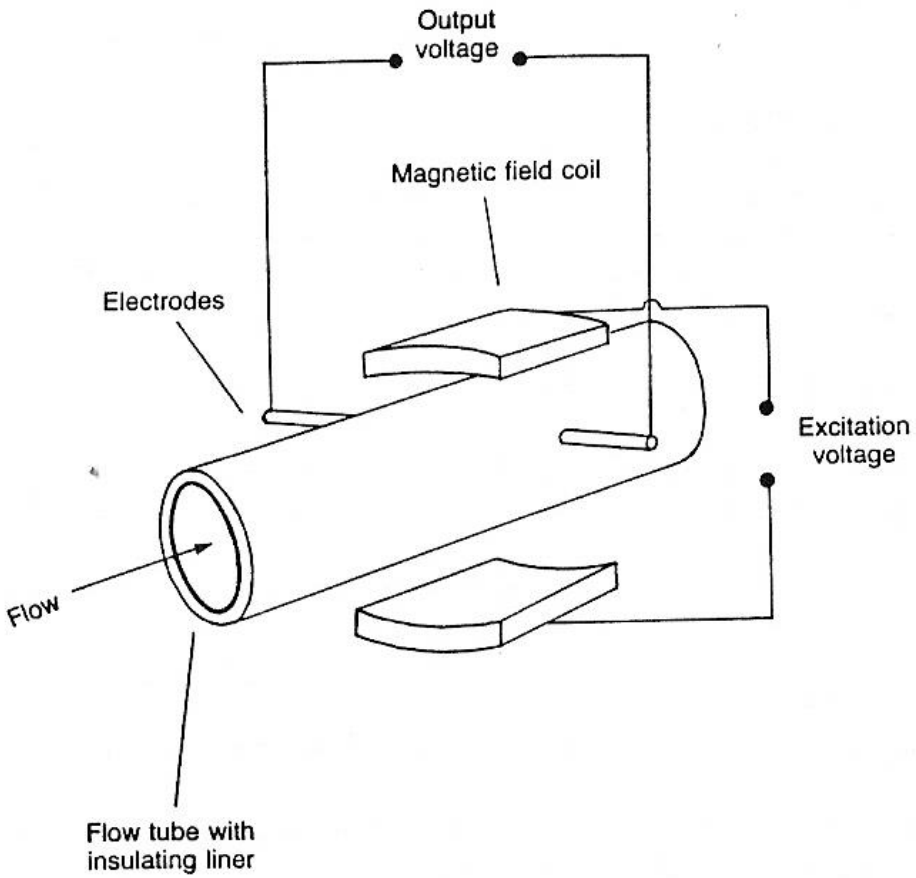


Flow Measurement Methods:
Electromagnetic Flow meters

- Can only work on conductive fluids

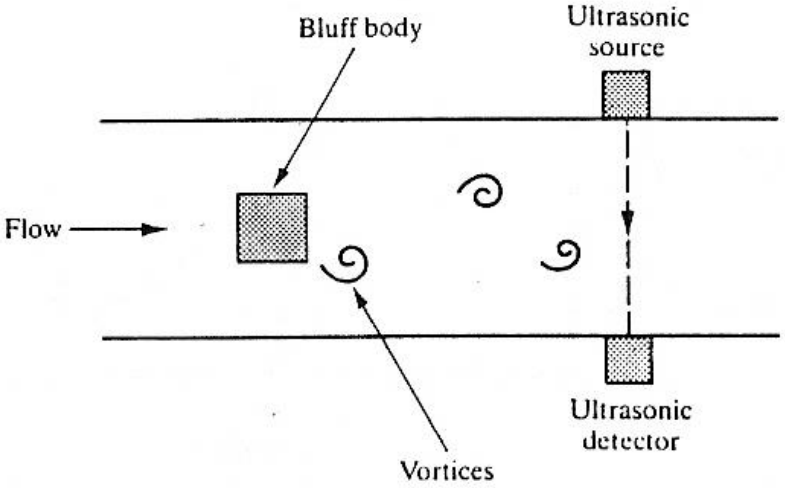
- Use Faraday's law of induction, which states that in the presences of a magnetic field *B*, a voltage *E* will be induced across a length *L* of liquid moving with a velocity *v* according to:

$$E = BLv$$



Flow Measurement Methods:
Vortex Shedding Flow meters

- Works according to the principle that an obstruction in a pipe will produce turbulence in a known fashion
- A centrally placed object, known as a bluff body, will shed vortices downstream in a very predictable and linear way
- An ultrasonic beam can be used to detect the vortices
- A pressure system can also be used to detect them



Flow Measurement Methods: Hot-Wire Anemometry

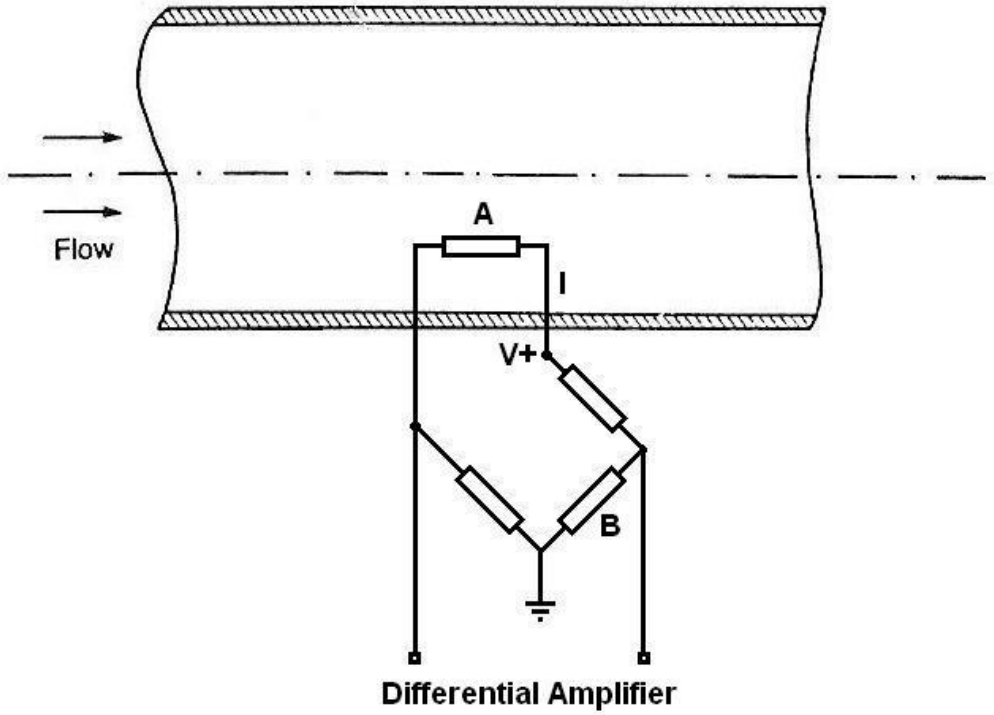
- This technique is widely used in industrial flow measurement
- The general principle is: A moving liquid will cool a hot object which it flows over and the faster it moves the cooler the object will be
- The way to implement it is to have two resistors, one of which is exposed to flow and one which is not
- Both resistors are identical and have identical currents flowing through them
- Each will be heated up by Ohmic heating

$$P = I^2 R$$

Flow Measurement Methods: Hot-Wire Anemometry

- The difference in temperature is measured by measuring the resistance of the resistors as all resistors have some known temperature coefficient

- A Wheatstone bridge arrangement can be used to detect changes in resistance



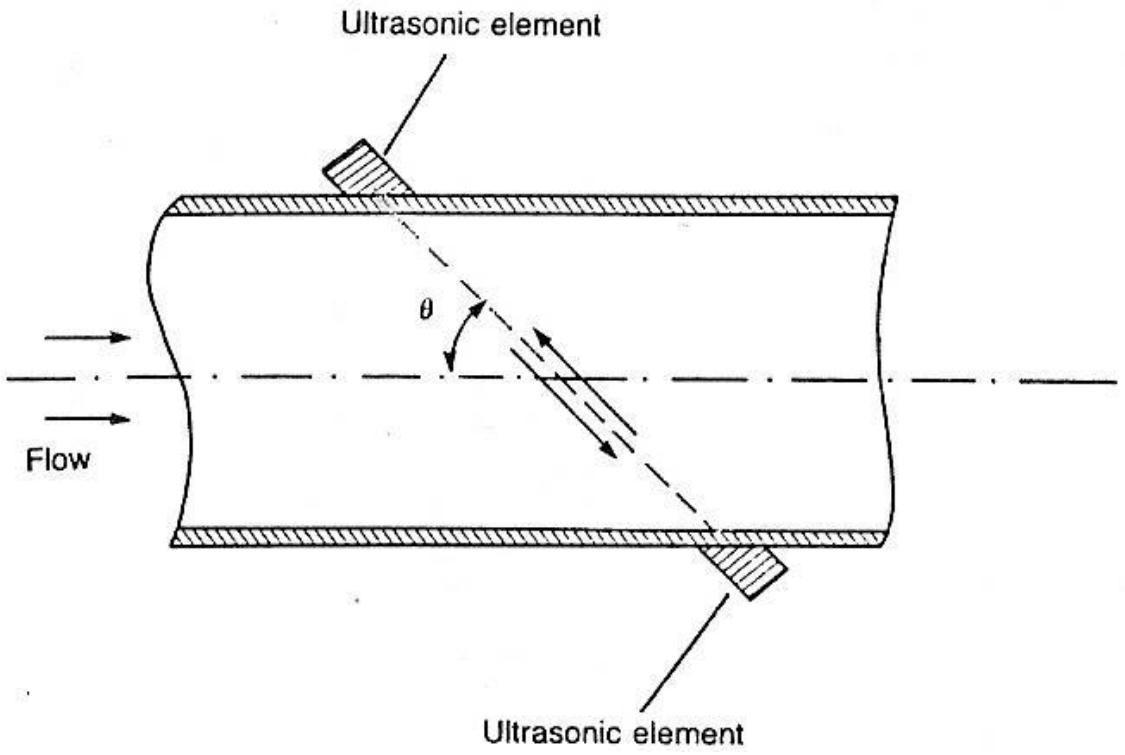
Flow Measurement Methods:
Ultrasonic Time-of-Flight Flow meters

- Sound does not have an absolute velocity as light does, it is sensitive to movements in the medium it is travelling in
- Sound which travels in a medium that is flowing in the opposite direction to the direction of the sound will travel more slowly than sound travelling in the same direction as the movement of the medium
- If we have an upstream and a downstream ultrasonic transmitter and receiver, we can measure the difference in the time of flight of the upstream and downstream signals
- From this we can calculate fluid flow velocity. The time difference is typically of the form

$$T = \frac{L}{\left(c \pm v\cos\theta\right)}$$

Flow Measurement Methods:

Ultrasonic Time-of-Flight Flow meters



Flow Measurement Methods:
Cross-Correlation Flow meters

- Work on the principle that there are features in the flow which can be detected at two different points

- For example, there maybe bubbles entrained in the flow or particles or pressure fluctuations

- If we can detect them, we can cross correlate them mathematically, which will give use the time difference for the travel between two points, according to the following relationship:

$$R(\tau) = \lim_{T \to \infty} \frac{1}{T} \int f(t) g(t + \tau) dt$$

- Where $f(t)$ and $g(t)$ are the detection signals at the two points. $R(\tau)$ will have a peak at the point where τ is equal to the time delay between the two points

Flow Measurement Methods:

Cross-Correlation Flow meters

